

HR Employee Attrition Analysis

End-to-End EDA, Statistical Testing & Predictive Modeling

1. Project Overview

This project analyzes the IBM HR Analytics Employee Attrition dataset (1,470 employees, 35 features) to identify the key drivers behind employee attrition. The workflow covers exploratory data analysis, hypothesis testing, feature engineering, and a logistic regression model to predict attrition. Overall attrition rate in the dataset is **16.1%** (237 of 1,470 employees).

2. Key Insights

2.1 Overtime — Strongest Individual Factor

Employees who work overtime show an attrition rate of **30.5%**, compared to **10.4%** for those who do not — nearly 3x higher. This was the strongest statistically significant relationship found (chi-square p-value = $8.15e-21$) and also had the highest correlation and regression coefficient among all features.

2.2 Job Level & Monthly Income

Attrition decreases sharply as job level rises: Level 1 employees show 26.3% attrition versus just 4.7% at Level 4 ($p = 6.63e-15$). This is closely tied to compensation — employees who left had a notably lower median monthly income (~3,202) compared to those who stayed (~5,204), a gap of roughly 38%. Since job level and income are highly correlated, job level appears to be the primary underlying driver.

2.3 Tenure — New Joiners at Highest Risk

Employees in their first 0-2 years at the company show a **28.9%** attrition rate, nearly double the overall average, and far higher than employees with 11-20 years tenure (6.7%). This effect is highly significant ($p = 1.66e-11$). Within the Sales department specifically, 34% of employees who left had 0-2 years of tenure — the largest single group of leavers.

2.4 Environment Satisfaction & Combined Satisfaction Score

Employees reporting the lowest environment satisfaction (level 1) show a 25.4% attrition rate versus the 16% overall average ($p = 5.12e-05$). Combining Environment and Job Satisfaction into a single average "Overall Satisfaction" score reveals an even clearer, near-monotonic trend: attrition falls from **37.7%** at the lowest combined score to **8.5%** at the highest — a more than 4x difference.

2.5 Department

Sales (20.6%) and Human Resources (19.0%) show notably higher attrition than Research & Development (13.8%), despite R&D being the largest department by headcount. This difference is statistically significant ($p = 0.0045$).

2.6 Distance From Home & Age

Employees living 21+ km from work show a higher attrition rate (~22%) than those living within 10 km (~14%), suggesting distance is a contributing factor. Employees who left also skew younger on average (mean age ~34) compared to those who stayed (~38), consistent with the tenure finding.

2.7 Factors Tested but Not Statistically Significant

Relationship Satisfaction ($p = 0.155$) and a derived Promotion Gap Ratio (Years Since Last Promotion / Years at Company, $p = 0.458$) showed no statistically significant relationship with attrition and were excluded from final conclusions, despite small differences appearing in raw percentages.

2.8 Summary Table

Factor	Higher-Risk Group	Lower-Risk Group	p-value
OverTime	30.5% (Yes)	10.4% (No)	8.15e-21
Job Level	26.3% (Level 1)	4.7% (Level 4)	6.63e-15
Tenure	28.9% (0-2 yrs)	6.7% (11-20 yrs)	1.66e-11
Environment Satisfaction	25.4% (Level 1)	16.0% (avg)	5.12e-05
Department	20.6% (Sales)	13.8% (R&D)	0.0045
Relationship Satisfaction	20.7% (Level 1)	15.0% (avg)	0.155 (n.s.)

3. Predictive Model — Logistic Regression

A logistic regression model was built to predict attrition using the full encoded feature set (categorical variables one-hot encoded, numeric features standardized). Given the class imbalance (84% No / 16% Yes), **class_weight='balanced'** was applied so the model does not simply default to predicting the majority class.

3.1 Model Performance

Metric	Class 0 (No)	Class 1 (Yes)	Overall
Precision	0.91	0.35	-
Recall	0.78	0.62	-
F1-score	0.84	0.44	-
Accuracy	-	-	75.2%

With class balancing, the model correctly identifies 62% of employees who actually left (recall), compared to only 19% before balancing — a meaningful improvement for the business goal of flagging at-risk employees, even though it comes at the cost of more false positives (precision of 0.35) and a lower overall accuracy (75% vs. 87% unbalanced). For attrition prediction, catching more true leavers is more valuable than raw accuracy.

3.2 Top Predictive Features

Overtime, frequent business travel, and job level ranked among the strongest coefficients in the model, consistent with the statistical testing results above. One caveat: Job Level and Monthly Income are highly correlated (multicollinearity), which can make individual coefficients for these two features unstable or counter-intuitive in sign — the univariate statistical tests (Section 2) remain the more reliable read on their individual effect.

4. Business Recommendations

- **Manage overtime load:** Overtime is the single strongest attrition driver (3x higher risk). Review workload distribution and overtime policies, especially in high-overtime teams.
- **Strengthen early-tenure retention:** Employees in their first two years are at the highest flight risk. Improve onboarding, early mentorship, and check-ins during this window — especially in Sales.
- **Review compensation at entry-level roles:** Level 1 employees show the highest attrition and lowest pay. Consider salary bands and growth paths for entry-level staff.
- **Focus on Sales and HR retention:** Both departments show attrition well above R&D; department-specific retention programs may help.
- **Monitor employee satisfaction proactively:** The combined satisfaction score showed one of the clearest, most consistent relationships with attrition — regular satisfaction pulse surveys could serve as an early-warning signal.

5. Tools & Methods Used

Python, Pandas, NumPy for data cleaning and analysis; Matplotlib and Seaborn for visualization; SciPy (chi-square tests) for statistical significance testing; Scikit-learn for logistic regression modeling.